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SARDAR PATEL INSTITUTE OF TECHNOLOGY
Munshi Nagar, Andheri (W), Mumbai – 400 058.
(Autonomous Institute Affiliated to University of Mumbai)

3.4: Research Publications and Awards

3.4.3: Number of research papers published per teacher in the Journals as notified on UGC CARE list during the last five years

Research Journal Paper published in Year Jan-Dec 2020

Sr. No	Research Paper Title	Author Names	Link to Papers
1	Analysis of Effectiveness of Quarantine Measures in Controlling COVID-19	Dr Dhananjay Kalbande, Dr.B.N. Chaudhari, Garima Kaushik, Shaney Mantri, Shrishti Kaushik	https://doi.org/10.1101/2020.04.21.20074245
2	Analysis of Therapy Transcripts using Natural Language Processing	Dhananjay Kalbande, Vrushali Joshi, Sarah Hawa, Shriya Akella, Shrishti Kaushik	https://www.ijeat.org/wp-content/uploads/papers/v9i6/F1598089620.pdf
3	AI Powered Holistic Solution for Travelers During Pandemic	Dhananjay Kalbande, Vedant Kumar, Siddhant Kumar, Parth Shingala, Pradhuman Singh,	https://zenodo.org/records/5530929
4	Feature Extraction Methods for Electroencephalography based Brain-Computer Interface: A Review	Sudhir Dhage and Dipti Pawar	http://www.iaeng.org/IJCS/issues_v47/issue_3/IJCS_47_3_18.pdf
5	Cross Language Application For Disease Prediction	Kiran Gawande, Shivani Bhutala, Niket Parekh, Advait Lad	http://thegrenze.com/index.php?display=page&view=journalabstract&absid=725&id=8
6	Road Pothole detection using Deep Learning classifiers	Surekha Arjapure, Dhananjay Kalbande	https://www.ijrte.org/wp-content/uploads/papers/v8i6/F7349038620.pdf

7	An artificial intelligence approach for the recognition of early stages of eczema	Dhananjay Kalbande, Rohit Naik, Janvi Jatakia, Uday Khopkar	https://www.inderscience.com/info/inarticle.php?artid=105656
8	Multiclass covert speech classification using extreme learning machine	Sudhir Dhage & Dipti Pawar	https://doi.org/10.1007/s13534-020-00152-x
9	An Overview of Speaker Recognition and Implementation of Speaker Diarization with Transcription	Arthav Mane, Janhavi Bhopale, Ria Motghare, Priya Chimurkar,	https://www.ijcaonline.org/archives/volume175/number31/mane-2020-ijca-920867.pdf
10	A Switchable Triple-Band Notched UWB Antenna Using Compact Multi-Via Electromagnetic Band Gap Structure	Vijay R. Kapure, Pramod P. Bhavarthe and S. S. Rathod	https://www.jpier.org/PIERC/pierc104/15.20052302.pdf
11	A metamaterial-inspired circularly polarized antenna for implantable applications	Siddhant Goswami, Deepak Karia	https://onlinelibrary.wiley.com/doi/full/10.1002/eng2.12251
12	Gas Geyser Syndrome: A cause that needs to be addressed immediately	Surendra Rathod, Manaswi Rananavare, Ria Shah, and Mansi Makwana	https://drive.google.com/file/d/1BodY7OYYy4xwZoRrNq_nzpf0068g8NZK/vi ew?usp=sharing
13	Instrumentation of Data Acquisition for EIT	Sukanya Aldar, Shruti Chavan, Dakshata Patil, Priya Chimurkar,	https://www.researchgate.net/publication/338632526_Instrumentation_of_Data_Acquisition_for_EIT
14	Application of Feature Selection and Random Forest Classifier on a Medical Dataset	Dr. Radha Shankarmani	http://ijste.org/Article.php?manuscript=IJSTEV6I10015
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22	MATE-Metro Assistance to an elderly App	Pallavi Thakur,Aftab sheikh, Hrishikesh Rajwade, Atif Kazi	https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4100953
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Analysis of Effectiveness of Quarantine Measures in Controlling COVID-19

 Garima Kaushik,  Shaney Mantri,  Shrishti Kaushik,  Dhananjay Kalbande, B. N. Chaudhari

doi: <https://doi.org/10.1101/2020.04.21.20074245>

This article is a preprint and has not been peer-reviewed [what does this mean?]. It reports new medical research that has yet to be evaluated and so should not be used to guide clinical practice.

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Abstract

COVID-19 has created an interesting discourse among the people of the world particularly regarding preventive measures of infectious diseases. In this paper, the authors forecast the spread of the Coronavirus outbreak and study how the reduction of transmission rates influences its decline. The paper makes use of the SIR (Susceptible Infected Recovered) Model which is a deterministic model used in the field of epidemiology-based on differential equations derived from sections of the population. The Basic Reproduction Number (R_0) represents the criticality of the epidemic in numeric terms. Forecasting an epidemic provides insights about the geographic spreading of the disease and the case incidences required to better inform intervention strategists about situations that may occur during the outbreak. Through this research paper, the authors wish to provide an insight into the impact of control measures on the pandemic. By drawing a comparison of three countries and their quarantine measures, observations on the decline of the outbreak are made. Authors intend to guide the intervention strategies of under-resourced countries like India and aid in the overall containment of the outbreak.

Competing Interest Statement

The authors have declared no competing interest.

Analysis of Therapy Transcripts using Natural Language Processing



Sarah Hawa, Shriya Akella, Shrishti Kaushik, Vrushali Joshi, Dhananjay Kalbande

Abstract: Mental health plays an integral part in leading a healthy life and having a positive outlook. This impacts our behavior, thought process, and actions and therefore it's important to identify and detect mental disorders in an early stage as it's effects can have a lasting influence on one's life. According to WHO, one in four people get affected by mental health disorders and currently 450 million people suffer from such conditions. Natural Language Processing can be a useful tool to analyze the trends in therapy transcripts. They can be further trained and optimized to derive useful insights and predict plausible future trends. Our proposed system analyses therapy transcripts and classifies it as 'Early signs of depression' and 'Serious after-effects of prolonged depression' based on the nature of the responses. Our system uses three different classifiers- Naïve Bayes, Support Vector Machine, and Logistic regression as well as two different vectorizers- TF-IDF and Count, to classify the text into these categories. This proposed system will not only help patients in identifying their symptoms but will also help therapists and researchers in gathering a large amount of data which could be used in predictive analysis, diagnosis and understanding the patient. Such research will pave the way for improving counselling and therapy sessions and be a very essential analysis tool for therapists.

Keywords: Naïve Bayes, TF-IDF, Vector Machine.

I. INTRODUCTION

Mental health denotes a person's state of psychological well being. A mentally healthy person is capable of utilizing their cognitive abilities to perform the usual functions that their body and surroundings demand. Mental illnesses can be treated through counselling, sometimes accompanied with prescribed medicine. According to WHO, one in four people get affected by mental health disorders and currently 450 million people suffer from such conditions [1]. Mental illnesses such as depression and anxiety can have dire consequences as they can lead to suicide, if not addressed and treated in due time. It becomes very crucial to identify such mental illnesses at an early stage. To make the process of identification efficient and quick, NLP based prediction can be utilised. With the help of a thorough analysis of therapy transcripts, early signs of mental illness such as depression, anxiety can be recognised.

Revised Manuscript Received on August 15, 2020.

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This analysis can further be used to obtain useful insights and predict future trends. Our proposed NLP algorithm analyses the transcripts of therapy sessions in order to perform classification based on the nature of responses. The data set consisting of therapy transcripts from counselling sessions is first analyzed to gain insight about the various characteristics of the transcripts and accordingly determine the features and scope of our algorithm. Then, the text is preprocessed in order to make it easy to understand for further processing. This is then followed by vectorization which is performed by calculation of TF IDF (Term frequency-inverse document frequency) and making use of Count vectorizer. This text is then classified into 'Early signs of depressions' and 'Serious after-effects of prolonged depression'.

II. LITERATURE SURVEY

Mental health has become exceedingly important to maintain a healthy lifestyle and a positive outlook. The awareness regarding mental health issues has gained prevalence in recent times. The system described in this paper talks about the classification performance of various classifiers such as Random Forests (RF), Support Vector Machines (SVM), Linear Discriminant Analysis (LDA), and k-Nearest Neighbor. LDA was considered the most efficient in terms of average generalization errors. One major drawback of LDA is that it only performs well on a smaller number of correlated features. [2] This paper monitors the Twitter feed of 135 Mechanical Turk (MTurk) participants and was compared with validated reports of suicide risk using a machine learning algorithm. The algorithm was able to identify the suicidal cases with 92% accuracy. The algorithm falls short in terms of identifying linguistic characteristics which is extremely vital in differentiating between various mental health-related causes. [3] A system that conducts emotion detection on Twitter feeds using natural language processing has been described in this paper. The tweets are classified as negative or positive using classifiers such as Support Vector Machine and Naïve Bayes. Multinomial Naïve Bayes model outperforms the Support Vector Machine model. A drawback of this system is the low accuracy caused by the use of short forms. [4] This paper talks about a system that processes Facebook data based on emotional process, temporal process, linguistic style, and all the aforementioned features together using techniques such as decision tree, k-Nearest Neighbor, Support Vector Machine, and ensemble. The Support Vector Machine model has better performance. However, the decision tree has better results for precision, recall, and f1 score. [5]



AI Powered Holistic Solution for Travelers during Pandemic



Vedant Kumar, Siddhant Kumar, Parth Shingala, Pradhuman Singh, Dhananjay Kalbande

Abstract: As the world is engulfed with COVID-19 pandemic and the glimpse of vaccine is still a distant dream, taking precautions and maintaining the norms suggested by WHO will keep us safe. With this, we present in this paper a solution that would help travelers induce confidence in traveling while keeping in mind the guidelines that must be followed. The solution focuses on an end to end service that will not only help the travelers to make informed and safe decisions but also allow the hospitality industry to monetize from this application. This paper is focused on a detailed analysis of the solution that is being presented to tackle the problems faced by various industries and their fear of resuming the work. A software-based approach is taken for providing a simple and engaging user experience to the user along with an AI approach to detect and predict the COVID trend in various cities. Along with this, a system that detects if people are wearing a mask or not will also be verified by the algorithm.

Keywords: Angular, Node JS, Machine Learning, Recurrent Neural Networks, Convolutional Neural Networks, MobileNet, ResNet

I. INTRODUCTION

It's been nearly a century since pandemics have globally affected not only the lives of the people but also the economy of many countries. It's proven that vaccines can take a lot of time to release. From previous pandemics like the plague of Justinian, Smallpox which wiped out a large number of populations, COVID too can have a disastrous effect on mankind if proper precautions are not taken into account. With COVID being a deadly pandemic, people are precarious to travel during and post COVID situation. It has called for following WHO norms like a compulsion of wearing masks. Apart from this, there is a prevalent uncertainty in the mind of people regarding the date of commute and travel. People are not sure about future trends and the situation of the place of travel. The main aim of this project is to induce confidence in people and to encourage them to travel along with helping the hospitality industry to function properly. For this, an easy to use the website is made for all the travelers so that they can be sure about the COVID trend and the news of COVID and other important updates of the place of travel. Using machine learning we can come up with a trend of high accuracy that predicts and plots the future COVID trend of the number of cases. Also,

by using Convolutional Neural Networks we achieved the task of detecting if people are wearing masks or not. This service can be used up by the hospitality industry like hotels and museums so that people visiting such places have an idea about the extent to which the WHO guidelines are being followed. This will enable travelers to make a safer decision about visiting a particular place for business or leisure

II. LITERATURE SURVEY

As per the reports by the World health organization (WHO) [1], it is evident that infections caused due to the corona virus disease 2019 (COVID-19) have crossed over 13 million people and the total deaths has crossed 580,000 globally. Furthermore, several other widespread detrimental respiratory diseases like the Middle East respiratory syndrome and severe acute respiratory syndrome have affected the global population and economy in a similar way. This has led to a rise in concern levels amongst people about their health. Public health has also been taken into consideration as the topmost priority for governments. Fortunately, works have shown that wearing face masks can prevent the spread of corona virus [2]. WHO has recommended that people who display respiratory symptoms, or the ones that are looking after symptomatic patients, should wear face masks [3]. Furthermore, strict guidelines have been issued and many public service providers are mandating the wearing of masks for the customers to use the service [4]. Therefore, the detection of face masks has become an imperative computer vision task to help society across the globe, but there exists limited research related to face mask detection. The underlying aim of the face mask detection process is to detect if a mask is being worn by a person or not along with detecting a face and providing a probability as an output [5]. Face detection, just like general object detection, detects an object of a particular class, i.e. face [6,7] and classifies it. Object and face detection have numerous applications that can be found in many domains, such as autonomous driving [8], education [9], surveillance, and many more. Standard object detectors usually rely on manually made feature extractors. Viola-Jones detector uses Haar features [10], while other famous methods include different feature extractors, such as scale-invariant feature transform, histogram of oriented gradients and many more [11]. Object detectors based on deep learning techniques have also demonstrated excellent performance and have surpassed standard object detectors. With the advent of Deep learning, neural networks can learn patterns and features relevant to a particular task [12]. There are two types of deep learning-based object detectors, namely, one-stage and two-stage.

Revised Manuscript Received on August 10, 2020.

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Feature Extraction Methods for Electroencephalography based Brain-Computer Interface: A Review

Dipti Pawar, *Member, IAENG*, and Sudhir Dhage

Abstract—Introduction: A brain-computer interface (BCI) is a rapidly growing cutting-edge technology in which a communication pathway is built between the human brain and computer. The BCI is also known as a direct neural interface where user can control external devices with the help of the brain signals. Neural signals are typically measured using electroencephalography (EEG).

Objective: Feature extraction from EEG data performs a significant role in the wearable BCI computing field. Since a large amount of EEG data, the major challenge is the effective feature extraction and reduce the computation burden. The objective of this paper is to review such different feature extraction techniques for the development of effective and robust BCI systems.

Approach: We reviewed feature extraction techniques employed in EEG based BCI studies. We synthesize these studies in order to present the taxonomy and report their usage with pros and cons.

Significance: This paper provides a comprehensive review of feature extraction techniques for EEG based BCI with their properties. Furthermore, open challenges are also discussed for further advancement in BCI studies.

Index Terms—Brain-Computer Interface, Electroencephalography (EEG), Feature Extraction, Brain-Machine Interface.

I. INTRODUCTION

Brain-Computer Interface (BCI) is a system which provides a direct communication pathway between the human brain and computers/machines by translating brain activity patterns into commands or messages for an interactive application. In general, brain activity patterns are measured using electroencephalography (EEG) which is noninvasive, easy to use and low cost a method. For instance, with the help of BCI it is possible to control computer without any physical activity and thus fundamentally useful in various applications e.g notably for motor impaired users to control assistive technologies by imagining the motor movement such as controlling wheelchairs [1], or controlling cursors [2], games [3], rehabilitation devices for stroke patients [4].

Figure 1 shows BCIs are as Pattern Recognition system. The system consists of two phases: 1) Training phase 2) Testing phase. The system is calibrated in the training phase while in the testing phase system recognize a neural pattern and translate them into useful commands for a device. Initially, EEG patterns are captured using the signal acquisition

system, which may also perform artefact processing and noise removal. Then these EEG signals are preprocessed using spectral and spatial features, followed by feature extraction technique which is compact representation of a signal. Lastly, these features are classified and classified signals translated into meaningful commands to control device.

The most difficult step in pattern recognition is still as it was stated some 65 years ago by [5], 'The extraction of significant features from a background of irrelevant details'. To date, very interesting BCI reviews are published including classification algorithm as in [6], [7]. However, none has been particularly dedicated to feature extraction techniques used for BCI and their properties. This study aims at filling this lack. Therefore one of the major objectives of this review is to study various feature extraction techniques utilized in BCI research and to analyse their critical properties. Another objective is to discuss some challenges which are important to solve considering future progress in BCI research.

The remainder of this review paper proceeds as follows. Section II describes the taxonomy of feature extraction methods used in EEG-based BCI. Section III reviews a wide range of feature extraction techniques for EEG based BCI. Section IV deals with discussion along with open challenges which are important for advancement in the BCI system. Finally, Section V concludes the paper.

II. TAXONOMY OF FEATURE EXTRACTION METHODS

EEG based BCI systems involve the extraction of useful information from the highly complex EEG data. In general, it is achieved by applying suitable feature extraction on EEG signals which are acquired by subjects during performing a specific mental activity. Subsequently, these features are fed to classifier for the training, where it gets learning to identify the pattern class. The EEG signals are highly nonstationary and dynamically changes due to technical and biological effects such as subject attention, the sessions variability, mental state, anatomical differences among subjects, amplifier and ambient noise [8]. Moreover, from the cognitive neuroscience perspective, the oscillatory (frequency) components of EEG have non-stationary and distinct characteristics. Due to these reasons, it is difficult to classify BCI patterns accurately. Therefore to enhance the performance of EEG based BCI systems, the selection of appropriate feature extraction technique is a very important issue.

The taxonomy of well established feature extraction methods that are applied to EEG is outlined in Figure 2. While there are many ways the two most commonly used feature extraction methods are time and frequency domain features.

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Cross Language Application for Disease Prediction

Journal: GRENZE International Journal of Engineering and Technology

Authors: Shivani Butala, Advait Lad, Niket Parekh, Kiran Gawande

Volume: 6

Issue: 2

Grenze ID: 01.GIJET.6.2.10

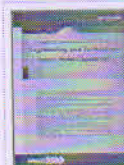
Pages: 114-122

Abstract

The infrequent and inadequate medical facilities for the people residing in the remote villages of India due to the lack of medical experts is a glaring medical issue facing the country. The proposed cross language application aims to solve the aforementioned problem with the use of technology by utilizing Translation techniques and Machine Learning models. With the usage of the proposed application, the users can vocally input their experienced symptoms into the application in their local language and get as output, the disease they possibly might be suffering from within seconds. To increase the accuracy of the results, an ensemble approach is followed in the paper which is a combination of five Machine Learning Models and a majority decision technique is applied to get the most accurate results. The proposed application is able to predict the disease with an accuracy of 90% in return of the symptoms provided by the user.

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TITLE:

GRENZE International Journal of Engineering and Technology

EDITOR IN CHIEF:

Dr. Janahanlal Stephen (Professor, Computer Science & Engg. Dept, Viswajyothi College of Engineering & Technology. Affiliated to KTU University, Kerala, S.India.)

ISSN:

2395-5295(Online); 2395-5287(Print)

FREQUENCY:

Twice Yearly

ASSOCIATED PUBLISHER:

GRENZE

RELATED SUBJECTS:

Computer Science & Engineering, Electrical & Electronics Engineering, Mechanical Engineering, Civil Engineering

RESEARCH GROUP:

GRENZE Engineering Group

INDEXING:

OFFICIAL WEBSITE:

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Road Pothole Detection using Deep Learning Classifiers



Surekha Arjapure, D. R. Kalbande

Abstract: Pothole is one of the major types of defects frequently found on the road whose assessment is necessary to process. It is one of the important reason of accidents on the road along with the wear and tear of vehicles. Road defects assessment is to be done through defects data collection and processing of this collected data. Currently, using various types of imaging systems data collection is near about becomes automated but an assessment of defects from collected data is still manual. Manual classification and evaluation of potholes are expensive, labour-intensive, time-consuming and thus slows down the overall road maintenance process. This paper describe a method for classification and detection of the potholes on road images using convolutional neural networks which are deep learning algorithms. In the proposed system we used convolutional neural networks based approach with pre-trained models to classify given input images into a pothole and non-pothole categories. The method was implemented in python using OpenCV library under windows and colab environment, trained on 722 and tested on 116 raw images. The results are evaluated and compared for convolutional neural networks and various seven pre-trained models through accuracy, precision and recall metrics. The results show that pre-trained models InceptionResNetV2 and DenseNet201 can detect potholes on road images with reasonably good accuracy of 89.66%.

Keywords: convolution neural networks , deep learning ,pothole detection, , pre-trained models

I. INTRODUCTION

The first step toward prompt repair of the road surface is to get known the condition of the road in terms of defects. The most common defect on road surfaces are potholes, bowl-shaped holes of various sizes in the pavement surface [1]. On road surface potholes are formed by a variety of reasons namely wear and tear of the road, age of the road, heavy rainfall, materials responsiveness to climate change, bad construction material used and external factors such as poor drain and quality construction management. Recently, damaged roads surface with potholes as shown in Fig. 1,

are increasing in India, and in Mumbai and thus complaints related to potholes are also increasing. Road with good condition always contributes a major portion to the country's economy.

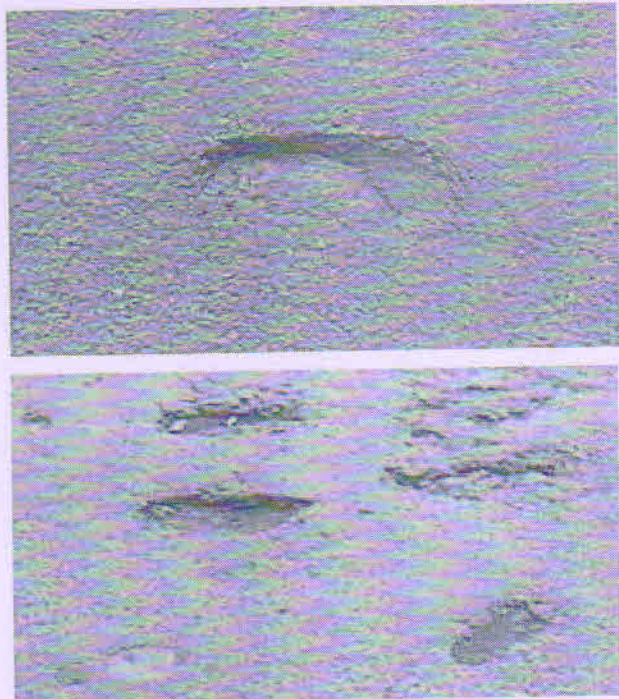


Fig. 1. Example of Potholes.

Effective maintenance of road surface is one of the major problems in developing countries becomes a great challenge for road maintenance authorities

The study of automatic distress analysis systems have obtained great attention in the past decades and many automatic distress analysis systems have been proposed. Road damages are important information for evaluating the road condition and conducting the necessary road maintenance activities. Conventional human visual pavement distress detection method is time-consuming, very expensive, labor-intensive and slows down the road maintenance management. Road surfaces get deteriorate over the age and thus they require corrective measures to restore safety and ride-ability. The need for fast accurate non-subjective inspection for road distresses is becoming important. The automated system is necessary to predict future deterioration rates and budgets. Pothole detection and estimation is one of the important tasks for the proper planning of reparation and rehabilitation of the road surface. At present, Road maintenance companies need many working hours for a rough estimation of damage on the road.

Manuscript received on February 10, 2020.
Revised Manuscript received on February 20, 2020.
Manuscript published on March 30, 2020.

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[International Journal of Medical Engineering and Informatics](#) > [2020 Vol.12 No.1](#)

Title: An artificial intelligence approach for the recognition of early stages of eczema

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Abstract: The rural population of India suffers from various medical ailments and due to the lack of medical facilities and practitioners, enough support is not available. Medical help might come late and the problem might have been aggravated. With the increasing awareness about artificial intelligence (AI), it is possible to solve these problems using technology. The research aims at detecting an early stage of the skin disease 'eczema', when the affected part of the human body is captured through a smart phone and approximate symptoms are provided by the medical practitioner. It uses artificial intelligence algorithms like convolutional neural networks and support vector machines algorithm for classifying the images, and back propagation algorithm for training a model based on the symptoms. Around 50 clinical photographs of eczema acquired from KEM Hospital, Mumbai to train the classifier and then different images of eczema were tested with an accuracy of greater than 85%.

Keywords: artificial intelligence; back propagation network; BPN; convolutional neural network; support vector machine.

DOI: [10.1504/IJMEI.2020.105656](#)

International Journal of Medical Engineering and Informatics, 2020 Vol.12 No.1, pp.52 - 61

Received: 09 Jun 2017

Accepted: 27 Apr 2018

Published online: 09 Mar 2020

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Original Article | [Published: 03 March 2020](#)

Multiclass covert speech classification using extreme learning machine

[Dipti Pawar](#)  & [Sudhir Dhage](#)

Biomedical Engineering Letters **10**, 217–226 (2020)

664 Accesses | **22** Citations | **1** Altmetric | [Metrics](#)

Abstract

The objective of the proposed research is to classify electroencephalography (EEG) data of covert speech words. Six subjects were asked to perform covert speech tasks i.e mental repetition of four different words i.e 'left', 'right', 'up' and 'down'.

Fifty trials for each word recorded for every subject. Kernel-based Extreme Learning Machine (kernel ELM) was used for multiclass and binary classification of EEG signals of covert speech words. We achieved a maximum multiclass and binary classification accuracy of (49.77%) and (85.57%) respectively. The kernel ELM achieves significantly higher accuracy compared to some of the most commonly used classification algorithms in Brain–Computer Interfaces (BCIs). Our findings suggested that covert speech EEG signals could be

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[Submitted on 10 Apr 2020]

An In-depth Walkthrough on Evolution of Neural Machine Translation

[Rohan Jagtap](#), [Dr. Sudhir N. Dhage](#)

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Neural Machine Translation (NMT) methodologies have burgeoned from using simple feed-forward architectures to the state of the art; viz. BERT model. The use cases of NMT models have been broadened from just language translations to conversational agents (chatbots), abstractive text summarization, image captioning, etc. which have proved to be a gem in their respective applications. This paper aims to study the major trends in Neural Machine Translation, the state of the art models in the domain and a high level comparison between them.

Comments: 10 pages, 10 figures

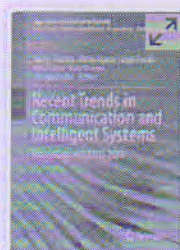
Subjects: **Computation and Language (cs.CL)**; Machine Learning (cs.LG); Neural and Evolutionary Computing (cs.NE)

Cite as: **arXiv:2004.04902 [cs.CL]**
(or **arXiv:2004.04902v1 [cs.CL]** for this version)
<https://doi.org/10.48550/arXiv.2004.04902>

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Recent Trends in Communication and Intelligent Systems pp 219–226

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Identification of Rumors on Twitter

[Richa Anant Patil](#) , [Kiran Gawande](#) & [Sudhir N. Dhage](#)

Conference paper | [First Online: 18 January 2020](#)

272 Accesses | **1** Citations

Part of the [Algorithms for Intelligent Systems](#) book series (AIS)

Abstract

Rumors are the statements about some incidents, peoples, things, event which are not true. They are harmful as it can hurt feelings, sentiments of people, it can spread misinformation, stress, anxiety, fear among the people. That is why accurate detection of rumors is necessary. Majority of the existing approaches are based on the semantic and syntactic analysis of tweet and user feedback. They have failed to detect the rumor if it is posted by a verified account. None of the approaches proposed so far paid attention to



Recent Trends in Communication and Intelligent Systems pp 133–141

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A Framework to Impede Phishing Activities Using Neural Net and Website Features

[Srushti S. Patil](#)  & [Sudhir N. Dhage](#)

Conference paper | [First Online: 18 January 2020](#)

245 Accesses

Part of the [Algorithms for Intelligent Systems](#) book series (AIS)

Abstract

Internet has metamorphosed into an unreliable platform due to the deceiving acts of phishing. With the emerging social networking platforms and growing online businesses, attackers find an enormous number of heedless victims. Envisaging this current scenario, the number of threats will keep on rising unless a gratifying solution is discovered. In this paper, we designed an anti-phishing framework to analyze a website using its


SPRINGER NATURE

An Overview of Speaker Recognition and Implementation of Speaker Diarization with Transcription

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ABSTRACT

This paper presents an overview of the generic process of a speaker recognition system and an implementation of its usage in a speaker diarization process. The motivation behind this paper is to present a simple implementation of a speaker diarization system that inculcates the usage of speaker recognition, speech segmentation and speech transcription. On the basis of various speech features such as Mel Frequency Cepstral Coefficients (MFCCs), Joint Factor Analysis (JFA), i-vectors, Probabilistic Linear Discriminant Analysis (PLDA), etc., speaker modelling is done to train Gaussian Mixture Models (GMMs), Hidden Markov Models (HMMs) and to use clustering. Speaker diarization is then implemented to get speakers speech segments which are then converted into text for the user. The methods discussed, and thus implemented, emphasize on maximum identification rate and minimal error in order to develop the functionality of speaker diarization and audio transcription and are aimed at helping the user to create a manuscript of the conversations that take place between multiple people.

Keywords

Speaker Recognition, Speaker Modelling, Audio Segmentation, Speaker Diarization, Speech Processing, Speech Recognition, Transcription

1. INTRODUCTION

Advancements in speech technology has attracted great minds of researchers and experts from all over the world. One part of this technology is the field of speech processing and speaker recognition. The extent of usage and scope of application for a robust speaker recognition system is unparalleled. Accounting to this, numerous studies have been published on how to implement a speaker recognition system, however, due its shortcomings in robustness, its usage is limited for applications such as biometric verification, speaker authentication, forensic purposes, etc.[1]. Nevertheless, the prominent reason behind the development of a

speaker recognition system is the barriers in speech processing for different speakers, each having a unique way of speaking, accent, pitch, rhythm, pronunciation, etc [2]. Research in this field has been going on for a long time, and consequently, studies have been published proposing the implementation of a speaker recognition system. For the development of an automatic speech recognition system, implementation of speaker identification modules followed by speech recognition modules have been proposed in previous studies [3].

The focus of this paper is to present an entire speaker recognition system as an application that can actually be used for automatically generating manuscripts of public meetings in groups, offices, classrooms, conferences, etc. Extending the usage of a speaker recognition system, use of speaker diarization [4] to fundamentally answer the question “which speaker spoke when?” in a conversation had in a meeting involving numerous speakers [5], is made. This is basically done by segmenting the conversation audio input into speech segments for different speakers based on the features extracted. This entire process of diarization [6] is implemented by basically applying the speaker recognition process coupled with unsupervised clustering. Furthermore, each of the segments obtained via diarization are transcribed to have the entire conversation in text format into a file for future reference.

2. BACKGROUND AND METHODS

A typical speaker recognition system is shown in Fig. 1. As seen in the figure, the entire system can be categorized into two types of processes, namely offline process and online process [1][2]. The offline process consists of creating a training algorithm for generating a background model based on the utterances of non-target speakers. The online process, on the other hand, involves the real-time processing of the speaker recognition system, using the trained models, for identifying the speaker in a new utterance. The first step involves training a base or background model, like Universal Background Model (UBM), after performing feature extraction on the utterances of non-target speakers. Once training

A Switchable Triple-Band Notched UWB Antenna Using Compact Multi-Via Electromagnetic Band Gap Structure

Vijay R. Kapure^{1,*}, Pramod P. Bhavarthe², and Surendra S. Rathod³

Abstract—In this paper, a miniaturized triple-band gap multi-via electromagnetic band gap (TBMV-EBG) structure is proposed. Lumped LC modelling method is used for the analysis of the proposed TBMV-EBG structure. Triple band gaps in both x and y -directions are obtained since TBMV-EBG unit cell consists of three resonance parallel LC circuits. Ansys (HFSS) simulation is used in eigen mode to simulate a unit cell of the proposed EBG. There is a strong agreement between simulated and experimental results. Comparing the proposed TBMV-EBG with triple band slotted EBG, triple band CSRR-EBG, fractal EBG, and dual band split EBG, size reductions of 6.52%, 7.53%, 23.21%, and 25.86% are obtained respectively which is validated by simulated and experimental results. Demonstration of the proposed TBMV-EBG structure for ultra-wideband (UWB) application is also presented. Simulation and measurement results prove that by using a single TBMV-EBG cell at the feed line of a UWB monopole antenna triple band-notches can be obtained. Moreover, switching characteristics of the proposed antenna are also demonstrated using single P-I-N diode. Depending on the ON and OFF switching status of P-I-N diode, the UWB antenna provides switching from triple band-notches to dual band-notches, respectively. The proposed switchable monopole UWB antenna as a single unit can be useful in applications wherein switching between multi-bands is desirable without changing the geometry of the structure.

1. INTRODUCTION

Band gap property of EBG structures plays an crucial role in microwave and antenna community due to their unique features. Periodicity of these EBG cells is realized by dielectric substrates metallic conductors like copper [1, 2]. Recently, many EBG structures with single/dual band gap characteristics have been reported. EBGs with triple band gaps are also required in some mobile applications like GSM900/DCS1800, global positioning system, WLAN (5.15 to 5.35 GHz, 5.725 to 5.825 GHz), and Worldwide interoperability for microwave access (WIMAX) [3, 4]. In [3], a triple band gap slotted EBG structure with four trapezoids and a co-joint small square is presented. The size of this EBG unit cell is 0.092λ at lower band gap center frequency of 1.72 GHz. In [4], a complementary split ring resonator (CSRR) is etched on conventional type of mushroom EBG to get triple bands in y -direction. The center frequencies of each band gap are 3.83 GHz, 5.67 GHz, and 9.49 GHz. The size of each EBG cell is 0.093λ at lower band gap center frequency of 3.82 GHz. In recent years, many compact dual-band EBG structures have also been reported. In [5], single EBG cell is used to get dual band-notches for UWB applications. In [6], a planar EBG design with square & circular loops is presented for radar cross section reduction (RCS) with the size of unit cell as 0.159λ at lower band gap center frequency of 3.40 GHz.

Received 23 May 2020, Accepted 29 July 2020, Scheduled 17 August 2020

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RESEARCH ARTICLE

WILEY

A metamaterial-inspired circularly polarized antenna for implantable applications

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Abstract

A novel microstrip-fed metamaterial-inspired antenna is proposed for implantable applications. The designed antenna operates in the Industrial, Scientific and Medical band at a frequency (2.4–2.5) GHz. The implantable antenna comprises of two split ring resonators. The use of metamaterial not only dramatically reduces the size of the antenna, but also improves its gain and bandwidth. Different methods of size reduction with optimum gain and bandwidth are here discussed in detail. The implantable antenna is designed using FR4 substrate which provides better gain, miniature size (84% size reduction), good impedance matching, circular polarization with 3.41% of 3-dB axial ratio bandwidth. Circularly polarized antennas are usually preferred as they reduce polarization mismatch loss. The antenna is $29 \times 28 \text{ mm}^2$ in size with 4.86 dB gain and 140 MHz impedance bandwidth. The design of phantom liquid, the computation of its pH value, and measurement of its permittivity are thoroughly reported. The permittivity of the liquid is found to increase with the pH value. The permittivity of phantom liquid has been validated with 85070E dielectric probe kit. A detailed description of different probes is provided. The measurement results are in good agreement with simulations. Finally, the design of phantom for implantable applications and techniques for size reduction of the same have been described. This reduced size will help to fit antenna in medical devices, providing greater ease for the body movement.

KEYWORDS

implantable antenna, metamaterial (split ring resonator), phantom design

1 | INTRODUCTION

Implantable antennas have gathered remarkable attention due to their importance in the medical and engineering field. These implantable devices are used for many kinds of medical applications such as tumor detection, wireless health monitoring, various cancer detection, speech sensing, self-monitoring, digestive monitoring,¹ retinal implants,² cardioverter defibrillators and pacemakers,³ temperature monitoring,⁴ and orthopedic monitoring device.^{5,6} Majorly it is reducing

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GAS GEYSER SYNDROME: A CAUSE THAT NEEDS TO BE ADDRESSED IMMEDIATELY

- Dr. S.S. Rathod, Ms. Manaswi Rananavare, Ms. Mansi Makwana, Ms. Ria Shah,
Sardar Patel Institute of Technology, Mumbai

Preface: This article is a modest attempt to bring into limelight all probable hazardous characteristics associated with gas geysers especially to be used for domestic purpose and keeping in mind the safety and security of the user. It also lists out different aspects of Gas Geyser related health Syndrome and how technological advancements can be used to provide effective solutions to this problem.

Introduction

There exists a dire necessity to create awareness on Carbon monoxide poisoning as well as associated neurological diseases arising out of accidental leak of vapors during the use of Gas based Geysers (water heaters) in bathrooms. They could prove to be fatal also as reported in recent times.

A resident of Borivali died on January 5, 2020. Prior to that (a week ago), he suffered severe neurological symptoms due to inhalation of carbon monoxide emanating from the gas geyser in the bathroom. Another case was found where a 25yr old person was rushed to the Emergency Room at Fortis Hospital, Mulund after experiencing dizziness and blackouts, Dr Sandeep Gore, Consultant and Head, Department of Emergency Medicine, Fortis Hospital, Mulund, signals that there has been a rise in cases of similar symptoms and nature, which is known as the 'Gas Geyser Syndrome'

What exactly is Gas Geyser Syndrome?

Liquefied petroleum gas (LPG) geysers are used very frequently for heating water in domestic holds. LPG gas is a combination of butane and propane. During the combustion of LPG gas, carbon dioxide and water are formed as byproducts. When the geyser is placed in an environment where there is no proper ventilation and the supply of oxygen is exiguous, incomplete combustion of gases inside the chamber occurs that releases hazardous carbon monoxide(CO).

Symptoms:

The symptoms include headache, nausea and fatigue. At higher concentration of CO, unconsciousness and death are possible.

What percentage of CO is harmful?

PPM	EFFECTS
0-125	No major symptoms observed.
200	Dizziness, nausea, fatigue & headache
400	Crucial 3 hours.
800	Unconscious, death within 2-3 hours.
1600	Death within 1-2 hours.
6400	Death within 30 mins.
12800	Death within 1-3 mins.

If a person feels unconscious, then First Aid: The victims should be immediately brought to a well-ventilated place and immediately seek medical help.

Instrumentation of Data Acquisition for EIT

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ABSTRACT

Electrical Impedance Tomography is a medical imaging technique which maps the resistivity distribution of the body and find the location of abnormalities present in the body part. In present techniques of imaging like CT scan, MRI and Ultrasound the patients have to move in specialized Room and bedside monitoring using these machines are not possible while it is possible by using EIT even in Intensive Care Unit where patients cannot be moved. This paper focuses on Instrumentation for Data acquisition for detection of location of cancerous tumors using Electrical Impedance Tomography. As we cannot experiment directly on the body, we are experimenting on phantom using saline water with adjacent current injection protocol having 16 electrodes attached on the periphery of Phantom. Our proposed research will be low cost, high speed and accurate solution for Data acquisition used in Imaging of EIT. This paper discussed on design of constant current source and Instrumentation for fast and accurate acquisition of voltages. Total 208 readings of voltages are taken by using switching circuit which is developed using Multiplexer, Demultiplexer, myRIO student embedded device and LabView.

General Terms

Electrical impedance tomography, LabView based advance instrumentation,

Keywords

Analog multiplexer, Electrical impedance tomography, LabView, myRIO, Instrumentation

1. INTRODUCTION

Electrical Impedance Tomography (EIT) is an imaging technique from two decades in which distribution of conductivity is calculated by placing electrodes on the surface of the body [1]. The objective of the EIT is to obtain the spatial distribution of the conductivity inside the human body and interpret it as an image [2].

Most commonly used methods for detection and monitoring of cancer are CT Scan, X-rays, MRI and Ultrasound. These techniques have impediments like harmful radiation which may cause side effects to the body [2]. Also, these machines are costly and not portable. Specialized room is required for diagnosis using these machines.

These all, limitations are taken care in EIT. In EIT, average differential voltage measurements with respect to adjacent current injection approach are used to form a tomographic image of tissues of human body. EIT does not lead to harmful radiation. Also, the magnitude of alternating current which is used as an input in this technique is very less (in mA) [3] due to which it won't cause any nerve stimulation. The operating frequency for alternating current is moderately high (10KHz

to 50KHz) so that it won't give rise to electrolytic effect in the human body [4]. Also, the ohmic power dissipation is less and it is diffused over the body which can be manageable by human thermo-regulatory system which makes EIT technique suitable for human body.

Due to the advantages of EIT like non-invasiveness, portability and radiation free, EIT is being studied and researched since 1970. Presently EIT is accepted as Gold method for the application in pulmonary monitoring, regional ventilation inside lungs and lung volume monitoring in hospitals. In this paper we are proposing EIT as a method for detection of type and location of Cancerous tumor in body [4]. There are two types of tumor, benign tumor which is analogous to non-conducting impurity while malignant tumor is analogous to conducting impurity [3].

Fig..01 is the block diagram of EIT system. Constant current is injected to the electrode pair and voltage from other pair of electrodes is acquired using switching circuit. Phantom having 16 electrodes attached to it is connected to switching circuit. For carrying switching operation and voltage acquisition between the electrodes myRIO and LabView is used.

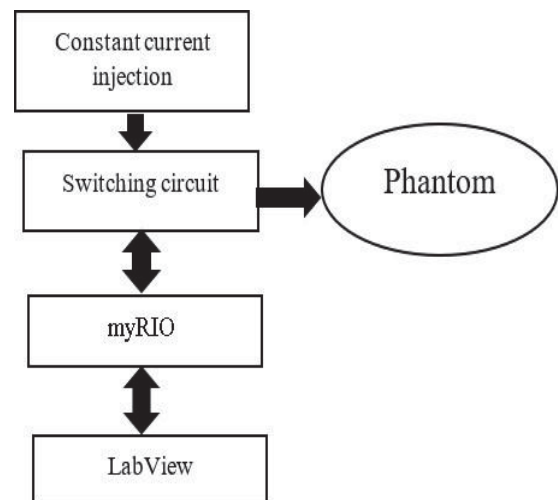


Fig.1 Block Diagram of EIT system

2. EIT SYSTEM

A circular plastic box of 15 cm diameter has been used to make a closed phantom. The ECG electrodes are placed around the circumference of box equidistant from each other at an angle of 22.5°. We have developed Automatic Data Acquisition system for Electrical Impedance Tomography using LabVIEW software, in which adjacent current injection approach is used for measurement. Fig.02 shows switching circuit of this system. 16 electrodes are placed at the surface of circular phantom. Alternating current as an input to a



ISSN (online) : 2349 - 784X

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ARCHIEVES

Application of Feature Selection and Random Forest Classifier on a Medical Dataset

Author(s):

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Keywords:

Feature Selection, Random Forest, Overfitting, Medical Dataset, Classification

Abstract:

For a given medical data set, there is a huge possibility that the data includes hundreds of features, each representing a symptom or a parameter based on which diagnosis can be carried out. While a lot of these features contribute towards the results, it is often the case that quite a few of these features turn out to be either irrelevant or have very

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M-JAW: Mobility-Based Jamming Avoidance in Wireless Sensor Networks

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Sudip Misra  ; Ayan Mondal  ; Prasenjit Bhavathankar ; Mohamed-Slim Alouini  [All Authors](#)

7

Paper

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Abstract

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[II. Related Work](#)

Abstract:

In this work, we study the problem of jamming avoidance for ensuring quality-of-service (QoS) in terms of the network lifetime and overhead in wireless sensor networks (WSNs). We propose a mobility model using Single-Leader-Multiple-Followers Stackelberg game theory to avoid the jamming affected region. In the proposed model, the centralized unit (CU) identifies the jamming affected region based on the locations of the affected nodes and acts as the leader. On the other

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International Journal of Computer Applications

Foundation of Computer Science (FCS), NY, USA

Volume 177 - Number 30

Year of Publication: 2020

Authors: Aashay Jain, Anukrit Jain, Yash Parekh, Renuka Pawar

[doi>](#) 10.5120/ijca2020919784

[Full Text](#)

Citation

Aashay Jain, Anukrit Jain, Yash Parekh and Renuka Pawar. Augment It Yourself. *International Journal of Computer Applications* 177(30):32-35, January 2020. [BibTeX](#)

Abstract

AR represents a way of exhibiting your product on a much more personalized level. It has the potential to invade spaces like a retail, restaurant, or event venue, and access customized and exclusively created AR content for you, which makes it a personalized event. Brands look for interactive technology to give customers a more personalized experience. AR acts as an entry point and lets brands access new markets and new customer base. Minutely created virtual 3D models, coupled with AR will

Bluetooth Based Electronic Notice Board.



Ashutosh Pandya, Chinmay Raut, Mihir Patel, Siddharth Das, Amol Deshpande

Abstract: Notice boards are of primary importance in any organization and in places such as bus and railway stations, when a need of for circulating notices arises it becomes tedious job. Thus an electronic notice board is an extremely efficient method of providing messages. It is difficult to update the messages at once. Thus this project focuses on development of a wireless board. This apparatus has the capability of displaying the latest messages using an Android application from a smart phone. This help susintransmitting any message with in a fraction of a second eliminating any delay by simply sending a command which is much efficient compared to any other traditional method of transmitting the message. Thus the proposed technology can be of great utility in many public places such as malls or commercial buildings to enhance the security system and also increase the awareness regarding emergency situations and avoid any possible dangers.

Keywords: Wireless, Microcontroller, GSM module, Bluetooth, Liquid crystal display, Rectifier, Regulator.

I. INTRODUCTION

Notice board is generally utilized in elementary schools and significant associations to pass on messages on the loose. A great deal of paper is been utilized and which just squandered. This prompts a great deal of deforestation that prompts a dangerous atmospheric deviation. Little creative decisions in utilizing innovation for ordinary purposes would positively affect the earth's issues which we are by and worried about. The crux of the venture is designing a structure of a notification driven scheduled notice Board that will supplant the current methods[1]. The fundamental target of the task is to develop a remote notification system that showcases any note which is transmitted by the client's android application gadget. Remote activity can be accomplished by any advanced cell/Tablet and so on, with a GUI (Graphical User Interface) based touch display. While the client sends the text from the mobile gadget and is recovered by a Bluetooth gadget on the receiver end[2]. The secret is known to the administrator then it can be delivered to the micro-controller that shows the notification sent from the client on the notice board which will be furnished with a crystal display. It utilizes an 8051 (SST89E51) to control its activity. Bluetooth will deal with the remote piece of the corresponding interface that is transmitted and get information between the two devices. While a mobile is currently in excess of a telephone nowadays, various

Applications are being based on a hagnoscope of stages for cell phone sisastounding. APC frame work tha fistulized for this reason for existing is known as a discourse synthesizer and can be actualized in both programming or equipment products. Wireless innovation has been gaining a huge ground over the previous decade. The consistently expanding utilization of remote systems today fills in as a marker of the advancement in the territory of remote networks. The interest in remote innovation is expanding in each aspect. The proposed framework in this report has numerous up and coming applications in instructive establishments and associations, wrongdoing avoidance, traffic the executives, railroads, promotions, and a lot more fields. Being easy to understand, working in along extend and quicker methods for passing on data are significant supports for this application. In other works done on a similar system that makes use of Zigbee and other short distance asynchronous communication channels. There is a detailed comparative study of different short-range wireless protocol which needs to be examined in order to select the one that will best suit the purpose[3], it was evident that the advantages of the system are less compared to that of the advantages. The main features including capacity, network topology, security, quality of service support, and power consumption are studied. Similar technology can be implemented using Zigbee [4], but it is not cost-effective By utilizing this proposed approach we can upgrade the security arrangement of wherever and make attention to the crisis circumstances and maintain a strategic distance from any potential danger. A remote printer is utilized for printing applications. The sound gadget is a speaker that is constrained by a microcontroller through the Text-To-Speech (TTS) converter [5] but that will increase the power consumption of the system and this is more of a problem for system.

II. BASIC PRINCIPLE

Taking into consideration, it tends to be inferred that, there exists a need for an electronic notice board that empowers a proficient direction to the client for showing notice. By thinking about increasing the smallness of electronic frameworks, there is a need of installing at least frameworks together. This venture is a usage of the possibility of remote correspondence among a mobile cellphone and an AVR controller. In this challenge work, we need to shape an inserted framework that incorporates display unit. The unit incorporates a show off that may be interfaced with microcontrollers. A remote printer is applied for printing applications. The sound device is a speaker this is constrained by a microcontroller via the Text-To-Speech (TTS) converter.

Revised Manuscript Received on September 25, 2020.

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SECURED IOT SOLUTIONS FOR HOME BASED AUTOMATION SYSTEM

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Abstract

This document discusses the security of home automation with the adaptation of Internet of Things. The Internet of Things (IOT) is a new boom in the technology world where physical things (home appliances, digital watches, etc.) are connected to the internet and there are new frameworks which give new ideas and imaginations to humans which are processed using cloud technology. With modern home designs, the living lifestyle has changed drastically. Brilliant home designs give an automation feel to the users to automate daily tasks. As technology has led dynamic change in the environment of household conditions, it has led to new security, validation and authentication challenges of this modern world. With all its advantages IOT also brings about some challenges which we need to tackle if we want efficient and safe use of IOT in home automation where we cannot risk the privacy and safety of our loved ones living in our homes.

Keywords —Home automation, Internet of Things, Security, CIAAA, Honey Encryption.

INTRODUCTION

Home automation grants you access to manage devices at your home from smartphones anywhere in the world. It's a stage towards what is referred to as the "Internet of Things," within which everything has a specific IP address, and can be monitored and accessed remotely. The primary and most evident recipients of this technique are smart devices and appliances that are connected to a local area network, via Ethernet or Wi-Fi. Be it, electrical frameworks and even individual points, similar to light switches and electrical outlets were additionally incorporated into home automation networks as they are fit for including and expanding a number of devices and systems.

On the basis of research, we consider five most important security goals (CIAAA) for smart homes which are as follows:

1. Confidentiality: It makes sure that only our family users can access the data.
2. Integrity: Data maintenance and modification is monitored so that outsiders are not allowed to spoof and alter the data.
3. Authentication: Verification between communication partners (user & receiver) of both parties is necessary.
4. Authorization: Access rights controlled system and the level of authority for children and elders are separated.
5. Availability: Any user should be able to access the data anywhere and anytime from mobile devices and online terminals.

This paper illustrates the Vulnerabilities to CIAAA and the challenges in IOT Home security and hence its forthcoming.

CONTAINERIZATION AND ITS EFFICIENCY ON
GOOGLE CLOUD PLATFORM□ Prof. Nikhita Mangaonkar*
Asha Arun Katale**
Priyanka Dilip Tikar***

ABSTRACT

So you want to move to 'Cloud'? You must have come across the term, 'Containerization' and how it is a big thing after 'Virtualization'. Technology trends are known to ride a wave, last decade was ruled by the term 'Virtualization', it allowed simulation of hardware functionalities to create multiple virtual computer systems on a single server, resulting benefits include economies of scale and greater efficiency. However, it also has serious drawbacks like performance degradation due to its heavyweight nature, lack of portability, lethargic provisioning, etc. Every drawback is a steppingstone, it is an innovation waiting to happen, enter 'Containerization'. It is a lightweight alternative to virtual machines. It a form of OS-level virtualization to deploy and run distributed applications without machines. This technology is currently surfboarding the Point Break wave, it is explosive, revolutionary and here to stay in a world where everything is moving to cloud. Containers allow us to package application together with its libraries and dependencies, into an isolated environment, that can be started anywhere, any number of times, much faster than their VM counterparts. Thus, containers and cloud have started walking together. Containers are agile, flexible and provide a standardized way to deploy an application without an operating system, they are fail-safe, easy to orchestrate and load balance on the go, a perfect fit for the cloud. Various cloud providers have jumped onto this bandwagon with their own offerings, each having its own unique advantages over others. All the offerings at the end of the day allow the developer to run some sort of containerized workload on their infrastructure, thus the term 'Container as a Service'. In this paper we'll implement such offerings from Google, a leader in this space, and see how it is wrapping containerization into various forms and providing it as a service to make lives easier, thus easing the complexities for organizations to move to the cloud.

Keywords—containerization, google-cloud, containers-as-a-service, Node.js

I. INTRODUCTION

Containerization allows bundling an application with all its configuration files, dependencies and libraries together into a single entity, that can spin-up or run on any container runtime ecosystem irrespective of the underlying infrastructure.

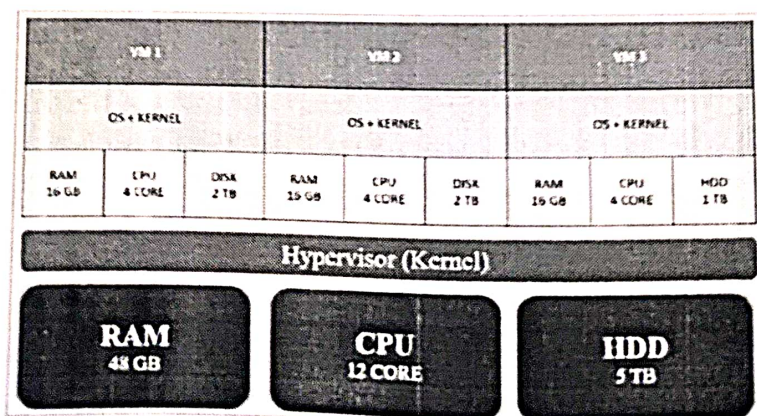


Figure 1: Virtual Machines - Architecture Diagram

**ANALYSIS OF GROUP DECISION-MAKING WITH OSCAR RESULTS
PREDICTION USING TWITTER API**□ Pranay Desai*
Prof. Pallavi Thakur****ABSTRACT**

Collaborative decision-making is a situation faced when individuals collectively make a choice from the alternatives before them. Twitter is American micro blogging and social networking service users post and interact with messages known as "tweets". The Academy Awards, more popularly known as the Oscars, are awards for artistic and technical merit in the film industry. This paper proposes a method to analysis results of Academy Awards based on Twitter data. The method extracts predictions from micro blogs made between announcement of nominees and award ceremony. The extracted data will be compared with the actual data to analysis the accuracy of prediction and effectiveness of Collaborative decision-making.

Index Terms—Oscars, twitter, collaborative decision-making, academy awards

I. INTRODUCTION

The Academy Awards, more popularly known as the Oscars, are awards for artistic and technical merit in the film industry. They release award nomination list ahead of the award show. Major News Agencies and people speculate the winner on Twitter by using hashtags. Twitter API we can evaluate how accurate News agencies and People where in predicting the results and refer it with results of Oscars 2020. The results will determine the accuracy of Collaborative decision-making.

**II. RELATED WORK ON PREDICTION
BASED ON TWITTER DATA ANALYSIS**

Predictions based on Twitter data has been studied in recently in current years, especially for predicting election results [1-7]. These predictions rely heavily on sentimental analysis data because twitter users don't usually announce their favourite candidate and their

decision can change at the last minute. Opinions on Motion picture are very strong compared to election results and Oscar speculations are posted on Twitter as soon as the final nomination list is announced by the Academy, hence the result generated will be more accurate. The aim with this data set is to analysis when voters participating in the decision-making process have less biases and adequate interest and knowledge about the topic, the decision made by them would more accurate[8].

III. METHODOLOGY

The initial analysis of tweet-data suggested to select English as the tweet language as English is the dominant tweet language and Oscars Awards are also conducted in English Language. The data is collected tweets, from 12 Jan 2020, when the academy announced the final nominees and 08 Feb 2020, day before the award ceremony through Twitter REST API. The data

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MATE- Metro Assistance To Elderly App

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Abstract—MATE app is an elderly assistance mobile application for providing help to the elderly on their fingertips. As the elderly comprise a major part of our population of our country their needs and preference are to be addressed. This research concentrates on monitoring elderly health providing help when necessary via a mobile application. The key aspect of the mobile application is to make the elderly feeling dependent and get help and care immediately whenever required. And as a human being becomes old generally he/she will experience health issues which could be very efficiently handled by this particular application. The application M.A.T.E.(Metro Assistance to the Elderly) provides custom help related to any health issue, with emergency equipment like wheelchair, stretcher. Also includes a feature to track location of the particular individual for the user's family, sanitary help, call helpline for guiding and solving the elderly's queries.

I. INTRODUCTION

The total Indian population achieves 1.35 billion according to the stats in 2017 out of which 104 million accounts to elderly people (above 60 years of age). According to some studies, India is ageing much faster than previously thought and may have nearly 20 per cent population of 60 years and above by 2050. Every 15 seconds, an older adult is admitted to the emergency room for a fall. A senior dies from falling every 29 minutes, making it the leading cause of injury among the elderly. Aging can bring about unique health issues. With seniors accounting for 12 percent of the world's population and rapidly increasing to over 22 percent by 2050 it's important to understand the challenges faced by people as they age and recognize that there are preventive measures that can place yourself on a path to healthy aging. Mate suggests a model of interaction between senior citizens and their surrounding environment while travelling through Metro trains. It helps to improve the independence living of elderly people and provides them a list of features that not only makes their travel convenient but also provides them an assistance, digital or physical as per their requirement. Mate is an android based application that helps us achieve all of this. This research aims to help senior citizens while travelling through metro trains not only in the train but also when they are in the vicinity of one.

Mate consists a pool of services that will make the travelling experience far less troublesome for the elderly people. It makes use of location tracking, secure login and profile creation for each user so Mate can help them more effectively. We discuss this revolutionary solution in detail in our research.

II. RELATED WORK

Senior citizens are often defined as people who are aged above 60-65 years [1]. It is predicted that by 2030 the number of elderly population (65 years and above) will reach 20 percent of the global population [1]. It has been observed that people of this age group face difficulties such as memory loss, hearing or vision problems, physical aid and other concerns. The efforts to improve healthcare for elderly people using mobile devices have been widely practiced for example Elderly that runs on android-os serves the ability to send alert such as sound and text, speed-dial contact setting, consumption of drugs schedule reminder and doctor appointment reminder on elder mobile phone remotely[1]. There is also AAL service which aims at helping elderly people when they are inside or near to some specific area as a bubble that includes public buildings and Inside the bubble the user can receive information and assistance related to the bubble. reproducing a beep, in short sentences and displayed with big letters to make easier the reading for elderly people [2]. Another application is iHelp, which is a real-time emergency reporting software dispatch system and rescue subsystem. It assists deaf-mute or elderly people in the shortest possible time in case of emergencies.[3].

III. LITERATURE REVIEW

The work presented in this paper has been developed in with the scope of aiding the elderly people. The idea of developing an application provides a vast coverage of audience through a medium which is widely accepted and used by almost all of the population [1]. The Mate app is designed to provides services to assist senior citizens having issues in walking or medical problems. Mate provides with information of the surroundings of the user and helps them to locate their position

Action Research on The Effectiveness of Autonomous Practices Upon Learning Outcomes

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Abstract: This paper presents a comparative study between the results obtained by two groups of students for a problem-based subject over two different systems of learning: the traditional, classroom-based delivery, and a system coupling that with a more interactive tutorial-based approach. The results show an improvement in the grade performance with the newer, interactive approach.

Keywords: Outcome based education, interactive teaching, action research

1. Introduction

The curriculum of an institution is a living entity. It evolves with time, reflecting the ever-changing needs of the society and keeps pace with the growing talent of the learners and the faculty. The objective of the curriculum is to develop professionals with competencies, intellectual skills and knowledge. These days the focus is on outcome based education (OBE), the purpose of which is to create a teaching and learning environment that would produce knowledgeable, creative and skilled professionals with positive values and attitude.

A. OBE Principles

An OBE curriculum gives a clear picture of what is important for learners to be able to do, and insights on how to then structure the curriculum, teaching learning and evaluation for effective learning. The four basic principles of outcome based education proposed by Spady (1994) are:

1) Clarity of focus: Everything that teachers do must be clearly directed and focused on what they want students to know, understand and be able to do. Teachers should focus on helping students to develop the knowledge, skills and personalities that will enable them to achieve the intended outcomes that have been clearly articulated.

2) Designing down: The curriculum design must start with a clear definition of the intended outcomes that students are to achieve by the end of the program. Once this has been done, all instructional decisions are then made to ensure achieving this desired end result.

3) High expectations: Teachers should establish high and challenging standards of performance in order to encourage students to engage deeply in what they are learning. Helping students to achieve high standards is linked very closely with the idea that successful learning promotes more successful learning.

4) Expanded opportunities: Teachers must strive to provide expanded opportunities for all students. This principle is based on the idea that not all learners can

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